

Safety Data Sheet(SDS)

1. Identification of the substance/mixture and of the company/undertaking

1) Product identifier : 2-ETHYL HEXYLACRYLATE (2-EHA)

2) Relevant identified uses of the substance or mixture and uses advised against

○ Relevant identified uses

Adhesives, sealants, Coatings and paints, thinners, paint removers, Polymer preparations and compounds

○ Uses advised against

Prohibited from use other than recommended

3) Supplier information

○ Company name [Manufacture]

Company : LG Chem, Ltd.

Address : 58, Yeosusandan 4-ro, Yeosu-si, Jeollanam-do, Republic of Korea

Emergency number : 080-880-0454

2. HAZARD IDENTIFICATION

1) Hazard classification

- Flammable liquids Category 4
- Skin corrosion/irritation Category 2
- Skin sensitization Category 1
- Carcinogenicity Category 2
- Specific target organ toxicity single exposure Category 3(Respiratory tract irritation)
- Hazardous to the aquatic environment, long-term (chronic) Chronic 3

2) Allocation label elements

Hazard pictograms



Signal word

- WARNING

Hazard statements

H227 Combustible liquid
H315 Causes skin irritation
H317 May cause an allergic skin reaction
H335 May cause respiratory irritation
H351 Suspected of causing cancer
H412 Harmful to aquatic life with long lasting effects

Precautionary statements

- Prevention

P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P261 Avoid breathing mist/vapours/spray.
P264 Wash skin thoroughly after handling.
P271 Use only outdoors or in a wellventilated area.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves/protective clothing/eye protection/face protection.

- Response

P302+P352 If you get on your skin: Wash with a large amount of water.
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P308+P313 If exposed or concerned: Get medical advice/attention.
P312 If you feel uncomfortable, receive medical institutions and doctors' consultation.
P321 When contacting with substances, make a treatment such as rinsing the skin and eyes in the water flowing more than 20 minutes.
P332+P313 If skin irritation occurs: Get medical advice/attention.
P333+P313 If skin irritation or a rash occurs: Get medical advice/attention.
P362+P364 Take off contaminated clothing and wash it before reuse.
P370+P378 In case of fire: Use carbon dioxide, Water, Alcoholic to extinguish.

- Storage

P403 Store in a wellventilated place.
P403+P233 Store in a wellventilated place. Keep container tightly closed.
P405 Store locked up.

- Disposal

P501 Dispose of contents and containers according to the legislation of the waste

3) Other hazards

- No data available

○ Product NFPA Level

Health	Flammability	Reactivity
2	2	2

(※ 0 = Stable , 1 = Low , 2 = Medium , 3 = High , 4 = Very High)

3. Composition/Information on ingredients

Components	Common name	CAS No.	PCT(wt%)
2-Ethylhexyl acrylate	2-Ethylhexyl acrylate	103-11-7	100

4. FIRST AID MEASURES

1) Following eye contact

- Seek immediate medical assistance.
- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.

2) Following skin contact

- For minor skin contact, avoid spreading material on unaffected skin.
- Seek immediate medical assistance.
- For hot product, immediately immerse in or flush the affected area with large amounts of cold water to dissipate heat.
- In case of burns, immediately cool the affected area with cold water for as long as possible and do not remove clothing that is stuck to the skin.
- In case of contact with substance, immediately flush skin or eyes with running water for at least 20 minutes.
- Wash skin with soap and water.
- Remove and isolate contaminated clothing and shoes.

3) Following inhalation

- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.
- Seek immediate medical assistance.
- Keep victim warm and quiet.
- Move to fresh air.
- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.
- Administer oxygen if breathing is difficult.
- Give artificial respiration if victim is not breathing.

4) Following ingestion

- Do not perform artificial respiration by mouth-to-mouth method, but use appropriate respiratory medical equipment.
- Seek immediate medical assistance.

5) Advice to physician

- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.

- Exposures require specialized first aid with contact and medical follow-up .

5. FIRE FIGHTING MEASURES

1) Suitable (and unsuitable) extinguishing media

- Suitable extinguishing media
 - CO₂.
 - Dry chemical.
 - For mixtures containing alcohol or polar solvent: Alcohol-resistant foam.
 - Water spray.
 - Use alcohol foam, carbon dioxide, or water spray when fighting fires involving this material.
 - Regular foam.
 - Use dry sand or earth to smother fire.
- Unsuitable extinguishing media
 - High-pressure water.
 - Direct water.

2) Special hazards arising from the substance or mixture

- Pyrolytic product
 - Can decompose at high temperatures forming toxic gases.
 - During burning, irritating and highly toxic gases may be produced by thermal decomposition or combustion.
- Risk of fire and explosion
 - Containers may explode when heated.
 - When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.
 - May violently polymerize and result in fire and explosion.
 - HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
 - Runoff may create fire or explosion hazard.
 - Vapor explosion hazard indoors, outdoors or in sewers.
 - Can form explosive mixtures at temperatures at or above the flashpoint.
 - Vapors may form explosive mixtures with air.
 - Vapors may travel to source of ignition and flash back.
- Other
 - Some liquids produce vapors that may cause dizziness or suffocation.

- Contact may cause burns to skin and eyes.
- Fire may produce irritating and/or toxic gases.

3) Special protective equipment for firefighters

- Rescuers should put on appropriate protective gear.
- Cautions ; Most of liquids are lighter than water.
- Most vapors are heavier than air. They will spread along ground and collect in low or confined areas (sewers, basements, tanks).
- Dike fire-control water for later disposal; do not scatter the material.
- Move containers from fire area if you can do it without risk.
- Evacuate area and fight fire from a safe distance.
- Fire involving Tanks: For massive fire, use unmanned hose holders or monitor nozzles; if this is impossible, withdraw from area and let fire burn.
- Fire involving Tanks: Cool containers with flooding quantities of water until well after fire is out.
- Fire involving Tanks: Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.
- Fire involving Tanks: Fight fire from maximum distance or use unmanned hose holders or monitor nozzles.
- Fire involving Tanks: ALWAYS stay away from tanks engulfed in fire.

6. ACCIDENTAL RELEASE MEASURES

1) Health considerations and protective equipment

- Do not touch or go near exposed material.
- The very fine particles can cause a fire or explosion, eliminate all ignition sources.
- ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area).
- All equipment used when handling the product must be grounded.
- Clean up spills immediately, observing precautions in Protective Equipment section.
- Stop leak if you can do it without risk.
- Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.
- A vapor suppressing foam may be used to reduce vapors.
- Cover with plastic sheet to prevent spreading.
- Please note that materials and conditions to be avoided.

2) Environmental precautions

- Runoff may cause pollution.
- Prevent entry into waterways, sewers, basements or confined areas.
- Keep out of waterways.

3) For cleaning up

- Absorb or cover with dry earth, sand or other non-combustible material and transfer to containers.
- Large Spill: Dike far ahead of liquid spill for later disposal.
- Absorb spill with inert material (e.g., dry sand or earth), then place in a chemical waste container.
- Small Spill: Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal.

- Dike and collect water used to fight fire.
- Absorb the liquid and scrub the area with detergent and water.
- Use clean non-sparking tools to collect absorbed material.

7. HANDLING AND STORAGE

1) Precautions for safe handling

- Avoid breathing vapors from heated material.
- Loosen closure cautiously before opening.
- CAUTION: High temperature.
- Handling refer to engineering control/personal protection section.
- All equipment used when handling the product must be grounded.
- DO NOT PRESSURIZE, CUT, WELD, BRAZE, SOLDER, DRILL, GRIND, OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION.
- Caution: Heat.
- Follow all SDS/label precautions even after container is emptied because they may retain product residues.
- Avoid prolonged or repeated contact with skin.
- Measure atmospheric oxygen concentration and ventilate the area during the operation since low-closed area can cause oxygen deficiency.
- Do not enter storage area unless adequately ventilated.
- Use care in handling/storage.
- Please note that materials and conditions to be avoided.
- Use only in a well-ventilated area.

2) Conditions for safe storage (including any incompatibilities)

- Empty drums should be completely drained, properly bunged, and promptly returned to a drum reconditioner, or properly disposed of.

8. EXPOSURE CONTROLS AND PERSONAL PROTECTION

1) Chemical exposure limits, Biological exposure standard

Components	Occupational exposure limits	ACGIH	Biological standard
2-Ethylhexyl acrylate	TWA : Not applicable STEL : Not applicable	TWA : Not applicable STEL : Not applicable	Not applicable

2) Appropriate engineering controls

- Use process enclosures, local exhaust ventilation, or other engineering controls to control airborne levels below recommended exposure limits.
- If user operations generate dust, fume, or mist, use ventilation to keep exposure to airborne contaminants below the exposure limit.
- Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

3) Personal protection equipment

- Respiratory protection
 - In case of insufficient oxygen (<19.6%), wear a supplied air mask or self-contained respirator.

- Wear breathing protection, which needs a confirmation from the Korea Occupational Safety and Health Agency.
- Eye protection
 - Wear suitable protective goggles and face shields.
- Hand protection
 - Wear suitable protective gloves.
- Body protection
 - Wear suitable protective clothing.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	transparent liquid
Physical state	Liquid
Colour	colorless
Odour	Characteristic odour
Odour threshold	0.02 mg/l
pH	No data available
Melting point/freezing point	-90 °C
Initial boiling point and boiling range	213.5°C
Flash point	86°C
Evaporation rate	No data available
Flammability(solid, gas)	Flammable liquid
Upper/lower flammability or explosive limits	6.4 / 0.8 %
Vapour pressure	0.178 mmHg (@ 25 °C)
Solubility(ies)	100 mg/l (@ 25 °C)
Vapour density	6.35
Relative density	0.89
n-octanol/water partition coefficient	3.67
Auto ignition temperature	252 °C
Decomposition temperature	No data available
Viscosity	1.38mm ² /s
Molecular weight(mass)	184.28

10. STABILITY AND REACTIVITY

1) Stability and hazardous reactivity

- Containers may explode when heated.
- When heated, vapors may form explosive mixtures with air: indoors, outdoors and sewers explosion hazards.
- May violently polymerize and result in fire and explosion.
- Can decompose at high temperatures forming toxic gases.
- HIGHLY FLAMMABLE: Will be easily ignited by heat, sparks or flames.
- Runoff may create fire or explosion hazard.
- Vapor explosion hazard indoors, outdoors or in sewers.
- Can form explosive mixtures at temperatures at or above the flashpoint.
- Some liquids produce vapors that may cause dizziness or suffocation.
- Contact may cause burns to skin and eyes.
- Vapors may form explosive mixtures with air.
- Fire may produce irritating and/or toxic gases.
- Fire may produce irritating, corrosive and/or toxic gases.

2) Conditions to avoid

- Containers may explode when heated.
- Ignition source(heat, spark, flame, etc.).
- Heat, contamination.
- Heat.
- Some liquids produce vapors that may cause dizziness or suffocation.
- Contact may cause burns to skin and eyes.
- Fire may produce irritating and/or toxic gases.

3) Incompatible materials

- Combustibles, reducing material.
- Containers may explode when heated.
- Some liquids produce vapors that may cause dizziness or suffocation.
- Contact may cause burns to skin and eyes.
- Fire may produce irritating and/or toxic gases.

4) Hazardous decomposition products

- Corrosive/toxic fume.
- Heat.
- Irritating and/or toxic gas.
- Irritating, corrosive and/or toxic gas.
- During burning, irritating and highly toxic gases may be produced by thermal decomposition or combustion.

11. TOXICOLOGICAL INFORMATION

1) Exposure route information

- Inhalation
 - May cause respiratory irritation
- Skin Contact
 - May cause an allergic skin reaction

- Causes skin irritation
- Eye Contact
 - Not applicable
- Ingestion
 - Not applicable

2) Health hazard information

- Acute toxicity
 - Acute toxicity(Oral)
LD50 4000 ~ 6000 mg/kg Test species: Rat, Source: OECD Screening Information Data Set
 - Acute toxicity(Dermal)
LD50 >10,000 mg/kg Test species: Rabbit, Source: OECD Screening Information Data Set
 - Acute toxicity(Inhalation:Gases)
No data available
 - Acute toxicity(Inhalation:Vapours)
No data available
 - Acute toxicity(Inhalation:Dust/mist)
No data available
- Skin corrosion/irritation
Severe irritation when exposed to rabbit skin, Source: OECD Screening Information Data Set
- Serious eye damage/eye irritation
Severe eye irritation in animal testing, Source: OECD Screening Information Data Set
- Respiratory sensitization
May cause skin sensitization, Mouse, OECD TG 429, Source : Ritu Chhimwal (2017)
- Skin sensitization
 - Positive reaction in human patch test - Moderate hypersensitivity in animal test, Source: OECD Screening Information Data Set
- Carcinogenicity
3 (IARC)
- Germ cell mutagenicity
Negative results in the in vitro bacterial mutation test - Negative results in the in vivo UDS assay [OECD TG 486], chromosomal aberration mutagenicity test, Source: OECD Screening Information Data Set
- Reproductive toxicity
No data available
- Specific target organ toxicity single exposure

An acute inhalation toxicity test in rats demonstrated nasal and eye irritation, a significant local irritant potential detected on the skin, and a slight irritant potential detected on the rabbit conjunctiva., Source : BASF AG (1958)

○ Specific target organ toxicity repeated exposure

In a 90-day rat repeated toxicity test, minimal liver damage was observed at 100 ppm, with high liver enzyme activity., Source: OECD Screening Information Data Set

○ Aspiration hazard

No data available

12. ECOLOGICAL INFORMATION

1) Aquatic toxicity

- Fish

LC50 1.8 mg/l 96 hr *Oncorhynchus mykiss*, Source: OECD Screening Information Data Set

- Crustacea

EC50 1.3 mg/l 48 hr *Daphnia magna*, Source: OECD Screening Information Data Set

- Aquatic algae

EC50 44 mg/l 72 hr Other (*Scenedesmus quadricauda*), Source: International Uniform Chemical Information Database(IUCLID)

2) Persistence and degradation

- n-octanol water partition coefficient

3.67 log Kow, Source: ICSC

- Degradation

No data available

- Biodegradation

(BOD(14 days) 51.7%), Source: National Institute of Technology and Evaluation(NITE)

3) Bioaccumulative potential

280, Source: National Library of Medicine/Hazardous Substances Data Bank(NLM/HSDB)

4) Mobility in soil

54954

5) Other adverse effects

No data available

13. DISPOSAL CONSIDERATIONS

1) Disposal methods

- Recyclables must be collected in portable bins and disposed of legally according to national regulations.

2) Precautions (including disposal of contaminated container of package)

- Avoid direct spillage to rivers, lakes, soil and drains.

14. TRANSPORT INFORMATION

- 1) UN No. : 3334
- 2) Proper shipping name : AVIATION REGULATED LIQUID, N.O.S.
- 3) Class or division : 9
- 4) Packing group : Not applicable
- 5) Marine pollutant : Not applicable
- 6) Special safety response for transportation or transportation measure :
 - Emergency measures in case of fire : Not applicable
 - Emergency measures in the effluent : Not applicable
- ADR
 - Tunnel restriction code : Not applicable
- IMDG
 - Marine pollutant : Not applicable
- Air transport(IATA)
 - UN No. : 3334
 - Proper shipping name : AVIATION REGULATED LIQUID, N.O.S.
 - Class or division : 9
 - Packing group : Not applicable

15. REGULATORY INFORMATION

- 1) Occupational Safety and Health Act in Korea
 - Not applicable
- 2) Toxic Chemical Control Act in Korea
 - Not applicable
- 3) Safety Control of Dangerous Substances Act in Korea
 - Class 4> Third petroleum (water-insoluble)(Designated quantity:2000ℓ)
- 4) Wastes Control Act in Korea
 - Not applicable
 - In case of disposal, it must be disposed of in accordance with Article 13 of the Waste Management Act.
- 5) Other regulations in KOREA and Abroad regulations
 - ETC regulation
 - Not applicable

- Not applicable
- Act on the registration and evaluation of chemicals
Existing Commercial Chemical Substances

16. OTHER INFORMATION

1) Reference

2) Date of first issue : 1997-01-20

3) Revision date

- Revised date count : 17
- Last revised date : 2025-06-30
- Last revised history : 2-EHA 한국 롤셋 MSDS 영문버전 개정

4) Other